

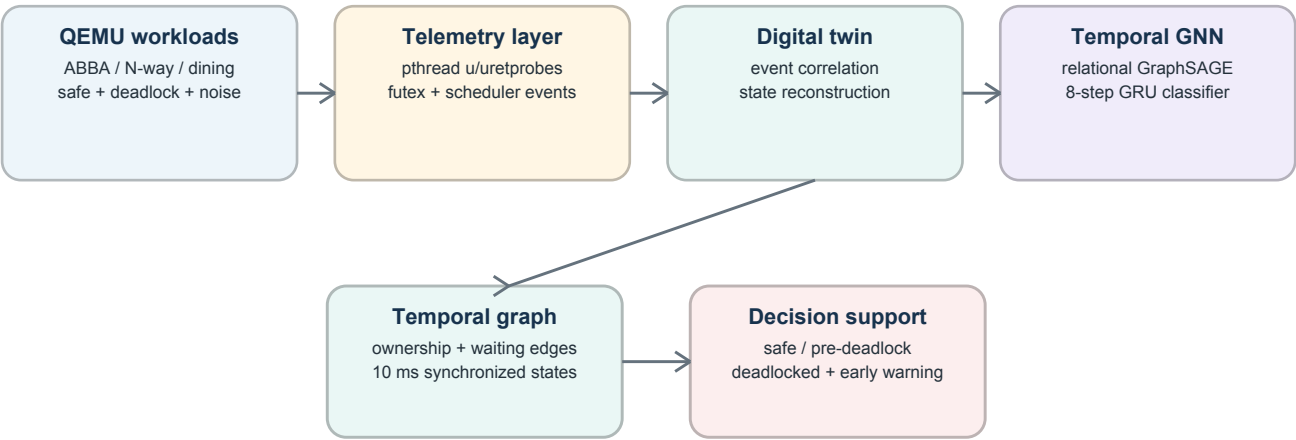
Temporal Synchronization Graph Dataset

Version 2 technical documentation

720	57,651	26,486	10 ms
independent QEMU runs	graph snapshots	temporal sequences	snapshot cadence

Purpose

A sequence-ready dataset for classifying Linux pthread synchronization states as safe, pre-deadlock, or deadlocked. Runtime events are collected inside an ARM64 Ubuntu QEMU guest and reconstructed as directed, heterogeneous thread-mutex graphs at a fixed 10 ms cadence.



Validated release: all 720 intended experiment configurations were selected, all workloads exited successfully, and the independent validator reported zero errors and zero warnings.

Release identity

Field	Value
Generation ID	v2-20260815
Guest environment	Ubuntu kernel 6.8.0-136-generic, aarch64
Processed size	149 MiB
Primary task	Three-class temporal graph classification

Experimental Design

The experiment matrix changes workload topology, execution seed, contention size, and background system activity. Each configuration is executed independently so telemetry from one run cannot contaminate another.

Dimension	Values	Coverage
Scenario	ABBA; N-way circular wait; dining philosophers	3 injectors
Mode	Safe; deadlock	Balanced by configuration
Threads	2, 4, 5, 8 (ABBA uses 2)	Multiple graph sizes
Noise	None; CPU; memory; mixed	4 host-load conditions
Seed	101-110	10 independent executions
Iterations	32 for safe workloads	Sustained contention

Split policy

Splits are globally seed-disjoint. A seed belongs to exactly one split across every scenario, thread count, mode, and noise condition. This is stricter than run-ID disjointness and reduces leakage from seed-specific scheduling patterns.

Split	Seeds	Runs	Snapshots	Sequences
Train	101-107	504	40,425	18,561
Validation	108	72	5,726	2,630
Test	109-110	144	11,500	5,295
Total	10 seeds	720	57,651	26,486

Sequence label distribution

Split	Safe	Pre-deadlock	Deadlocked	Total
Train	10,903	4,382	3,276	18,561
Validation	1,527	635	468	2,630
Test	3,075	1,284	936	5,295

Recommended training class weights: safe 0.5675, pre-deadlock 1.4119, deadlocked 1.8886. These inverse-frequency weights are stored in `metadata.json`.

Telemetry and Digital-Twin Reconstruction

pthread mutex operations often complete entirely in user space. Futex tracing alone therefore misses uncontended lock acquisition and release. The collector combines user-space probes with kernel-visible blocking and scheduler activity.

Sensor	Role in state reconstruction
pthread mutex uprobes / uretprobes	Observe lock, trylock and unlock attempts and successful returns, including uncontended user-space operations.
Futex syscall tracepoints	Identify threads that enter and return from kernel blocking waits.
Scheduler switch and wakeup	Capture scheduling activity, wakeups, last CPU and migration behavior.
Thread/process exit	Remove stale thread state and close lifecycle boundaries.
Workload markers	Provide controlled ground truth and the scenario mutex allowlist.

Temporal Synchronization Graph

Element	Representation	Meaning
Thread node	thread:<pid>:<tid>	Linux thread participating in the observed synchronization state.
Lock node	lock:<pid>:<address>	Scenario mutex/futex address in one process.
Ownership edge	lock -> thread	Successful acquisition; relation type owned_by.
Waiting edge	thread -> lock	Blocked waiter; relation type waits_for.

Features available to the model

Thread nodes expose waiting state, wait duration, scheduler switches, wakeups, CPU migrations and last observed CPU. Lock nodes expose ownership state. Labels, cycle flags and nodes, label metadata, provenance, run/snapshot IDs, and run-level performance aggregates are excluded from the smoke-model feature tensors.

Every sequence references eight consecutive snapshots from one run. Graphs are stored once in each split JSONL; sequence files contain compact ordered snapshot-ID references with a stride of two snapshots.

Label Semantics and Quality Controls

Class	Operational definition	Decision meaning
Safe	No observed cycle and outside the 300 ms pre-cycle horizon.	Continue monitoring.
Pre-deadlock	Developing state within 300 ms before first cycle, or during the 50 ms confirmation interval.	Generate early warning.
Deadlocked	Circular wait persists for at least 50 ms.	Generate deadlock alert.

Acceptance gates

The builder rejects a deadlock run unless all three classes occur and at least one pre-deadlock snapshot is strictly earlier than the first observed cycle. Deadlock runs are retained for 300 ms after first cycle. Safe runs must contain safe labels only.

Validation category	Checks performed	Result
Completeness	All expected configuration keys; unique run IDs; successful workload status	Pass
Graph structure	Node/edge references, relation directions, type invariants, finite numeric features	Pass
Temporal integrity	10 ms deltas, monotonic timestamps, consecutive sequence indices, no cross-run windows	Pass
Label integrity	All classes per split, safe/deadlock semantics, true pre-cycle warning states	Pass
Leakage	Disjoint run IDs and globally disjoint seed sets	Pass
Sensors	Required mutex, futex and scheduler coverage per applicable run	Pass

Final validation: 720 runs, 57,651 snapshots and 26,486 sequences; zero errors and zero warnings. Every run contained telemetry, with observed minima of 16 lock attempts, 14 acquisitions and 11 scheduler switches.

Topology diversity

The validator found 340 unique topology-label signatures in training, 199 in validation and 256 in test. The largest single signature accounts for approximately 6.3% of each split, limiting dominance by one repeated graph pattern.

Trainability Evidence

A relational GraphSAGE encoder processes directed incoming and outgoing ownership and waiting relations. Mean-pooled thread and lock embeddings are passed through a GRU over eight graph snapshots, followed by a three-class classifier.

0.213	0.072	98.75%	33.33%
epoch 1 loss	epoch 4 loss	validation macro-F1	majority baseline

Smoke-test setting	Value
Training sample	3,000 balanced sequences (1,000 per class)
Validation sample	1,200 balanced sequences (400 per class), seed 108 only
Epoch losses	0.2129, 0.1030, 0.0813, 0.0723
Validation accuracy / macro-F1	0.9875 / 0.9875
Gradient check	Finite for all optimization steps

This result demonstrates that the files, features, sequence references and neural training path are usable. It is a controlled smoke test, not evidence of production accuracy. The held-out test split was not used in this check.

Reproduction

```
python3 scripts/qemu/run_matrix.py --config config/experiments.v2.json --resume
PYTHONPATH=src python3 -m deadlock_dataset.cli temporal-build --runs-dir runs --output dataset/v2 --config
config/experiments.v2.json
PYTHONPATH=src python3 -m deadlock_dataset.cli temporal-validate --dataset dataset/v2
.venv/bin/python scripts/train_temporal_smoke.py --dataset dataset/v2 --output
dataset/v2/trainability_report.json
```

Limitations

The dataset is synthetic and controlled. It covers pthread mutex/futex behavior, not arbitrary production software, other synchronization primitives, distributed deadlocks, or kernel-internal lock deadlocks. Strong model results should be followed by independent evaluation on real applications and physical Linux hosts.

Primary release artifacts

dataset/v2/metadata.json - measured composition and split policy
dataset/v2/validation_report.json - validator output
dataset/v2/trainability_report.json - smoke-test metrics
dataset/v2/run_manifest.jsonl - per-run configuration and sensor counts